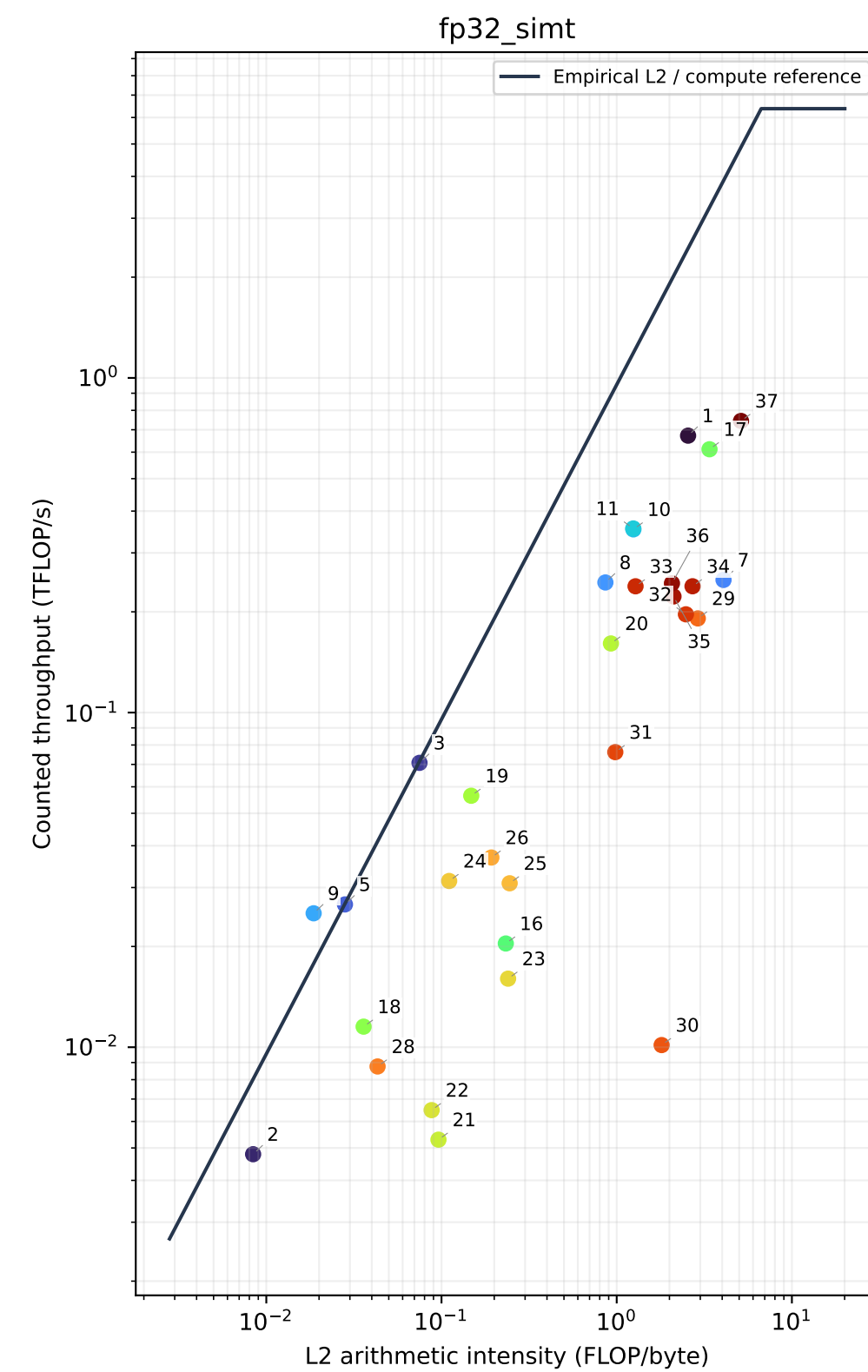
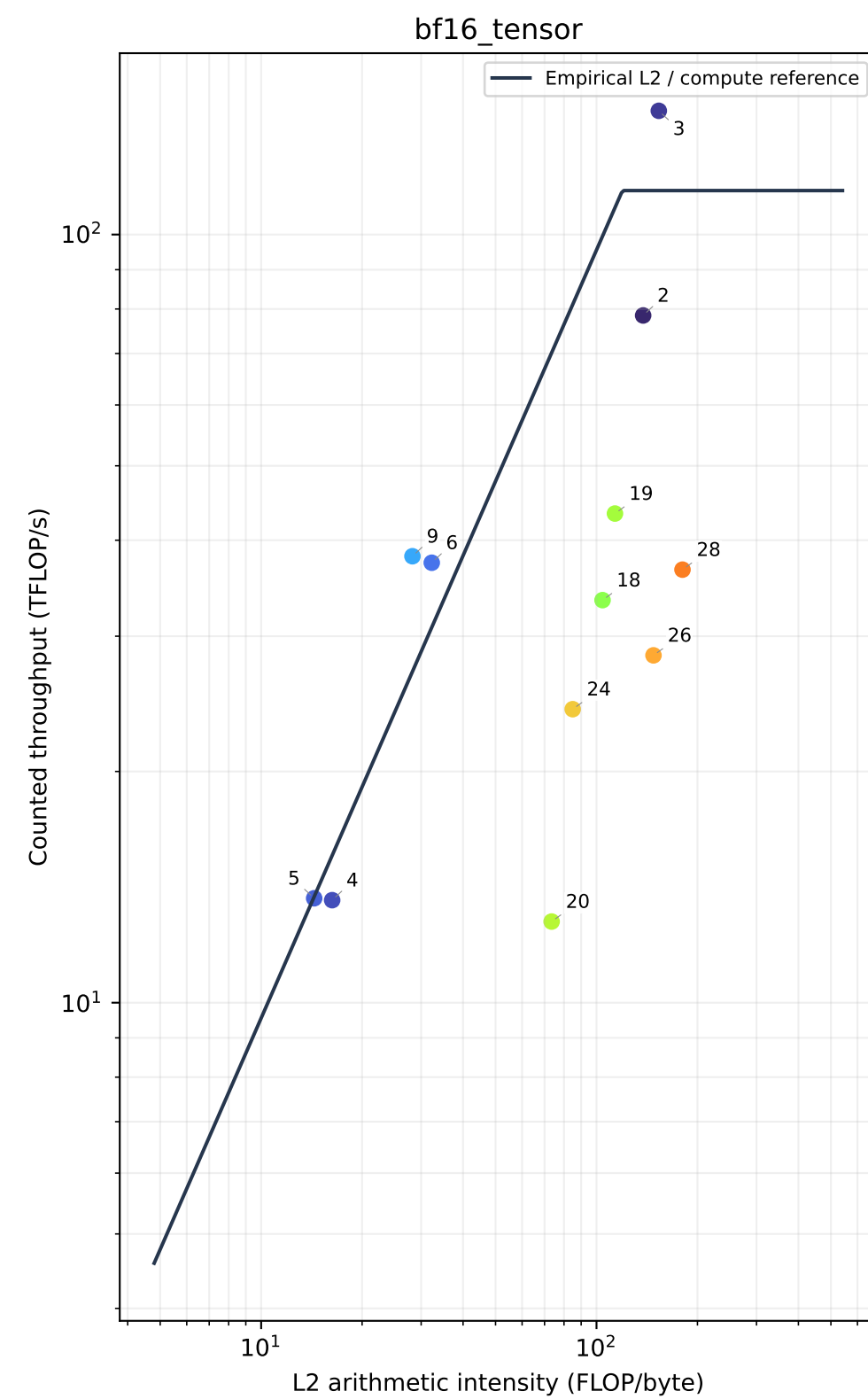


π 0.5 VLM — roofline by operator



POINT → OPERATOR

- 1 Prefix FFN: GELU product
- 2 Prefix FFN: down + residual
- 3 Prefix FFN: gate/up projection
- 4 Prefix: Attention $P \times V$
- 5 Prefix: Attention QK^T + scale
- 6 Prefix: Attention output + residual
- 7 Prefix: Prefix RoPE table
- 8 Prefix: Prefix masked softmax
- 9 Prefix: QKV projection + Gemma RoPE
- 10 Prefix: RMSNorm (pre-FFN)
- 11 Prefix: RMSNorm (pre-attention)
- 12 Prefix: concatenate image/text embeddings [no counted FLOPs]
- 13 Prefix: mask conversion/copy [no counted FLOPs]
- 14 Prefix: valid-length fill [no counted FLOPs]
- 15 Text: embedding lookup [no counted FLOPs]
- 16 Text: embedding scale
- 17 Vision: FFN GELU
- 18 Vision: FFN contraction + bias
- 19 Vision: FFN expansion + bias
- 20 Vision: FlashAttention
- 21 Vision: NCHW → NHWC (input)
- 22 Vision: NCHW → NHWC (weights)
- 23 Vision: NHWC → NCHW
- 24 Vision: attention Q/K/V/O projections
- 25 Vision: image embedding scale
- 26 Vision: image-to-text projection + bias
- 27 Vision: input cast/layout [no counted FLOPs]
- 28 Vision: patch embedding convolution
- 29 Vision: patch/LayerNorm fusion 1
- 30 Vision: patch/LayerNorm fusion 2
- 31 Vision: patch/LayerNorm fusion 3
- 32 Vision: residual/LayerNorm fusion 10
- 33 Vision: residual/LayerNorm fusion 4
- 34 Vision: residual/LayerNorm fusion 6
- 35 Vision: residual/LayerNorm fusion 7
- 36 Vision: residual/LayerNorm fusion 8
- 37 Vision: residual/LayerNorm fusion 9